

the diffuser inlet. Greater losses generated by fluctuations of the approach flow velocity at the diffuser inlet are the consequence.

4. The vane angles at the diffuser inlet can be calculated according to Eqs. (3.18 and 3.19). The vane blockage is generally low but can be considered in c_{3m}' .
5. To limit the pressure losses in the column pipe and the outlet bend, the flow velocity in the column pipe should not exceed 5 m/s. The annular cross section at the diffuser outlet must be dimensioned accordingly.
6. Since the design aims at a swirl-free discharge flow with $\alpha_4 = 90^\circ$ and the flow in the diffuser cannot be vane-congruent, the vane angle should be selected at $\alpha_{4B} = 94$ to 96° .
7. The vanes can be designed as arcs of circles. The leading edges can be rounded or shaped as elliptical profiles. The trailing edge is designed blunt (rounding on the outlet would be unfavorable because of the formation of vortex streets, Table 10.13).

7.7 Inducers

To reduce the required $NPSH_R$ of a pump it is possible to install an inducer upstream of the actual impeller. As an example, Fig. 2.1 shows the section of a single-stage process pump with inducer. An inducer typically allows a reduction in the value of the required $NPSH_R$ of a pump to half of the value without inducer.

The inducer increases the static pressure upstream of the impeller. Cavitation bubble development on the impeller blades is reduced or suppressed accordingly. As a consequence, the pump can be operated at a higher speed or a lower $NPSH_A$ than without an inducer, provided that the inducer does require significantly less $NPSH_R$ than the impeller.

Inducers for industrial application typically reach suction specific speeds of $n_{ss} = 400$ to 700 , Chap. 6.2.4. These high suction specific speeds are achieved by small approach flow angles β_{1a} (refer to Fig. 6.21), thin leading edge profiles with low blade blockage and long channels in which the cavitation bubbles generated in the inducer implode at least partially before the fluid enters the actual impeller. The low approach flow angle and the small flow coefficient ϕ_1 at the inducer inlet require a considerably enlarged inlet area compared with the impeller. This is achieved by reducing the hub ratio and by enlarging the inlet diameter.

The recommendations and data for the inducer design discussed in the following largely originate from work done by the NASA published in [19, 29, 30, 45], and in [B.16, B.24]. The information is tailored for industrial applications with suction specific speeds of up to $n_{ss} \approx 600$. Inducers with far higher suction specific speeds are employed in the aerospace technology where even two-stage inducers are used at times. The design methods discussed in the following do not fundamentally differ from those of the high-performance inducers.

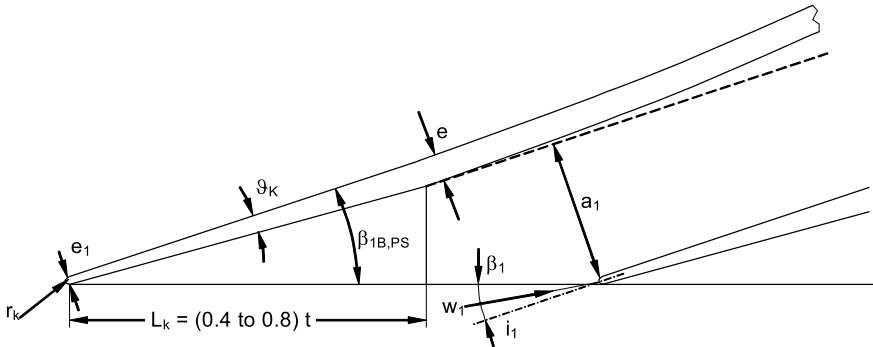


Fig. 7.33 Selection of the blade geometry at the inducer inlet

As is evident from the example in Fig. 2.1, the flow in the inducer is essentially axial. Often the inlet diameter is slightly larger than that of the downstream impeller. The hub diameter at the inducer inlet is designed as small as possible so that the hub contour is generally conical with the hub radius growing towards the impeller inlet. As with axial impellers, the static pressure in the inducer is increased in that the relative velocity is decelerated, Eq. (T3.3.8). This must be considered when deciding on one of two possible blading concepts:

- Blades with constant pitch designed as screw surfaces with blade angles $\beta_{1B} = \beta(L) = \beta_{2B} = \text{constant}$ on cylindrical sections. These blades only generate pressure when the incidence $i_1 = \beta_{1B} - \beta_1$ is greater than zero or for as long as the flow in the blade channel is decelerated according to $w_{1q} < w_1$. The advantage of this blading is its simple manufacture.
- Blades with variable pitch (Fig. 7.34, right) where the angle on cylinder sections grows from the inlet to the outlet as with a propeller pump: $\beta_{2B} > \beta(L) > \beta_{1B}$.

7.7.1 Calculation of Inducer Parameters

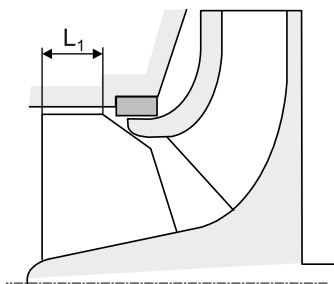
Determination of main dimensions and inlet conditions, Table 7.6

1. **Basis:** Known parameters are the pump selected for the relevant application, its speed n , flow rate Q_{opt} , NPSH_R curve and the dimensions of the impeller. In addition, the suction specific speed to be achieved by the inducer or its NPSH_R is specified. Furthermore, the boundary conditions for the mechanical design must be ascertained. Usually the approach flow angle is $\alpha_1 = 90^\circ$ and the axial velocity is assumed as constant over the inlet cross section.
2. **Design flow rate:** Like axial impellers, inducers have a steep Q-H-curve. The pressure rise therefore drops rapidly to zero above the shockless entry (depending on the type). For this reason, the maximum required flow rate of the

pump must be taken into account when determining the design flow rate of the inducer by selecting $Q_{\text{ind}} = (1.1 \text{ to } 1.15) \times Q_{\text{opt}}$ or by making use of Eq. (7.6.7).

3. **Blade number $z_{\text{VL}a}$:** Inducers usually have 2 to 4 blades.
4. **Inducer inlet diameter d_1 :** The inlet is designed for the selected suction specific speed. The optimum flow coefficient for the required suction specific speed can be selected from Eq. (T7.6.3). The appropriate inlet diameter can then be determined. Alternatively, the optimum inlet diameter can be calculated for selected coefficients λ_w and λ_c from Eq. (T7.6.1). The expected suction specific speed is obtained from Eq. (T7.6.4). Finally, it is also possible to calculate $\phi_{1,\text{opt}}$ from Eq. (T7.6.2). All of these options yield the same result. For inducers with axial inlet $\lambda_c = 1$ to 1.1 and $\lambda_w = 0.02$ to 0.08 can be assumed. The inlet diameter must also be matched to the inlet nozzle and the downstream impeller, which can imply considerable restrictions on the design options in case of standard pumps.

Iteration may be required with respect to the hub ratio. Two uncertainties must be considered in this calculation: (a) As discussed in Chap. 6.3.2, the coefficients λ_c and λ_w depend on the geometry; the values assumed in the calculation will be attained only if the blade design is as good as in the measurements from which the coefficients had been derived. (b) The NPSH_R of the pump is not only governed by the inducer but also by the downstream impeller. If the impeller has not been specifically designed for operation with the inducer, the design suction specific speed of the inducer may not be reached under certain conditions, Chap. 7.7.3. This risk is particularly high in the range of large flow rates ($q^* > 1$); it grows with increasing suction specific speed of the inducer.



Arranging an inducer in the meridional section



Fig. 7.34 Impeller with variable-pitch inducer

Table 7.6 Inducer design, $\alpha_1 = 90^\circ$ (Fig. 7.33)			Eq.
Optimum inlet diameter for given λ_w	$\lambda_c = 1.1$ $\lambda_w = 0.02$ to 0.08	$d_{1,opt} = 3.25 \left(\frac{Q_{La}}{n k_n} \right)^{\frac{1}{3}} \left(\frac{\lambda_c + \lambda_w}{\lambda_w} \right)^{\frac{1}{6}}$	7.6.1
Optimum flow coefficient for given λ_w	$\phi_{1,opt} = 0.1$ to 0.18	$\phi_{1,opt} = \sqrt{\frac{\lambda_w}{2(\lambda_c + \lambda_w)}}$	7.6.2
Optimum flow coefficient for given n_{ss} $n_{ss,Ref} = 400$	$\phi_{1,opt} = 0.15 \left(\frac{n_{ss,Ref}}{n_{ss}} \right)^{0.93}$	$d_{1,opt} = 2.9 \left(\frac{Q_{opt}}{k_n n \tan \beta_1} \right)^{\frac{1}{3}}$	7.6.3
Suction specific speed $n_{ss} = 350$ to 700	$\phi_1 = \begin{cases} n_{ss,Ref} \\ n_{ss} \end{cases}^{0.93}$ $n_{ss,Ref} = 52$	$n_{ss} = \frac{98}{(\lambda_c + \lambda_w)^{0.25}} \left(\frac{k_n}{\lambda_w} \right)^{0.5}$	7.6.4
Flow inlet angle at outer streamline	$\beta_{1,a,opt} = \arctan \phi_{1,opt}$	$\beta_{1,a,opt} = \arctan \frac{240 Q_{opt}}{\pi^2 d_{1,opt}^3 n k_n}$	7.6.5
Pressure surface blade inlet angle at outer streamline	Criterion: design incidence: $i_{1,opt} = (0.35 \text{ to } 0.45) \times \beta_{1B,PS}$	$\beta_{1B,a,DS} = (1.5 \text{ to } 1.8) \times \beta_{1,a,opt}$	7.6.6
$i_1 = 2$ to 4°	Criterion: incidence at maximum flow rate: $i_{1,Qmax} = 1^\circ$	$\beta_{1B,a,DS} - 1^\circ \geq \beta_{1,max} = \arctan \frac{240 Q_{max}}{\pi^2 d_1^3 n k_n}$	7.6.7
Blade inlet angles over blade height; same incidence for all streamlines		$\tan \beta_{1B,DS}(r) = \frac{r_a}{r} \tan \beta_{1B,a,DS}$	7.6.8
Blade leading edge profile (s. Fig. 7.33)		$\vartheta_k = (0.35 \text{ to } 0.5) \times \beta_{1B,a,DS}$	7.6.9
Check also: $i_1 \geq \vartheta_k + 0.5^\circ$		$L_k = (0.4 \text{ to } 0.8) \times t$	
		$r_k = (0.03 \text{ to } 0.05) \times e$	
Cavitation coefficient only for $i_1 > 0$ $k_3 = 0.22$ to 0.65	$\sigma_3 = k_3 k_n \left\{ \frac{\phi_1}{\sqrt{k_n}} - \frac{\phi_1^2}{\tan \beta_{1B}} \left(1 - 0.36 \frac{k_n}{\tau_1} \right) + \frac{3.18 \times 10^{-3}}{\tau_1 \left(\tan \beta_{1B} - \phi_1 \tau_1 \sqrt{k_n} \right)} \right\}$		7.6.10
Influence of tip clearance on n_{ss}	$\frac{n_{ss}}{n_{ss,s=0}} = 1 - k_4 \sqrt{\frac{s}{b_1}}$	$k_4 = 0.5$ to 0.65	7.6.11
Blade length		$(L/t)_a = 1$ to 2.5	7.6.12
Hydraulic efficiency		$\eta_h = 1 - 0.11 \times L/t$	7.6.13
Deviation angle δ_2 Flow angle $\beta_2 = \beta_{2B} - \delta_2$		$\delta_2 = \left(2 + \frac{\beta_{2B} - \beta_{1B}}{3} \right) \left(\frac{t}{L} \right)^{1/3}$	7.6.14
Static pressure rise in inducer. The calculation is done for the mean streamline		$\psi_p = \eta_h \left[1 - \phi_1^2 \left\{ \frac{A_1^2}{A_2^2 \sin^2(\beta_{2B} - \delta_2)} - 1 \right\} \right]$	7.6.15
Note: Use <i>pressure surface</i> blade angle $\beta_{1B,PS}$ rather than camber angle.			
Number of inducer blades: $z_{Ind} = 2$ to 4 . Sweep-back $\epsilon_{sb} = 65$ to 90° according to Fig. 7.37			
Hub diameter (given by mechanical design)	Inlet: Select hub diameter close to minimum: $v_{1,min} \approx 0.15$ Outlet: Match to impeller hub; a large hub diameter helps to increase the head developed by the inducer (centrifugal forces)		

5. **Inlet diameter at the inner streamline d_{1i} :** The lower limit for d_{1i} is obtained from a minimum hub ratio of $v_{1,\min} \approx 0.15$. Smaller values of d_{1i} are unfavorable since excessive blade inlet angles would result at the inner streamline.
6. **Blade inlet angle β_{1B} :** When the inducer inlet diameter has been determined according to step (4), all quantities of the inlet velocity triangle according to Table 3.1 can be calculated. The first section of the inducer blades is designed according to Fig. 7.33. The calculation is based on the angle on the pressure surface of the blade $\beta_{1B} = \beta_{1B,PS}$ which is obtained from the flow angle plus an incidence of $i_1 = 2$ to 4° from Eqs. (T7.6.5 to 7.6.7). In this way, the flow and blade angles are calculated for outer, mean and inner streamline. When determining the inlet angles, it must be observed that the steep rise of the $NPSH_R$ -curve (depending on the blade configuration) occurs near the flow rate of shockless entry. In addition to the design point it is therefore necessary to consider the maximum required flow rate at which the incidence should not drop below one degree to avoid cavitation on the pressure surface of the blades, Eq. (T7.6.7).
7. **Inlet throat area A_{1q} :** As with impellers (see Chap. 7.2.1, item 15), the deceleration of the relative velocity vector w_1 to the mean flow velocity w_{1q} in the throat area must be limited in order to avoid premature inlet recirculation. Conversely, the fluid must not be accelerated at the maximum flow rate with respect to overload cavitation. The throat area at the inducer inlet must therefore be selected so that the ratio w_{1q}/w_1 in the design point is in the range of 0.6 to 0.75 but does not exceed the value $w_{1q}/w_1 = 1$ at the intended maximum flow rate.
8. **Inducer outlet diameter $d_{2,a}$ and $d_{2,i}$:** The diameters on the outer streamline and on the hub are obtained from the dimensions of the downstream impeller. A conical hub of the inducer with $d_{2i} > d_{1i}$ is favorable since the head is increased through centrifugal forces and since the flow on the hub requires less severe deflection with increasing d_{2i} (lower tendency to stall). If the inducer diameter at the outlet is smaller than at the inlet, $d_{2,a} < d_1$, the first cylindrical part (i.e. dimension L_1 in Fig. 7.34) must be designed sufficiently long as to cover at least the throat area of the inducer.

Relationships at the outlet of the inducer: The blade outlet angle must be chosen so that the static pressure in front of the impeller increases sufficiently in the specified operating range to cover the NPSH requirement of the impeller. The effect of the static pressure rise in the inducer on the NPSH requirement of the system inducer/impeller is explained in Fig. 7.35 which was calculated on the basis of measurements on a process pump. The measured $NPSH_{3,imp}$ of the impeller alone was plotted against the flow rate, curve 1.

The static pressure rise H_p in the inducer is plotted as curve 3. Curve 3 passes through zero at Q_B . The inducer operates as a pump at $Q < Q_B$, but as a brake at $Q > Q_B$, (Chap. 12). If the static pressure rise in the inducer (curve 3) is subtracted from the $NPSH_{3,imp}$ of the impeller (curve 1: without inducer), curve 5 is obtained which passes through zero at Q_u . In the domain with $Q < Q_u$, (that is at

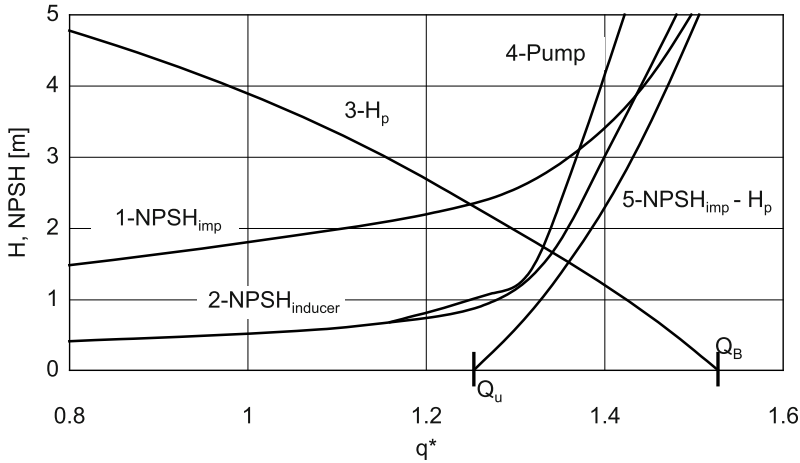


Fig. 7.35 Inducer characteristics

$H_p > \text{NPSH}_{3,\text{imp}}$) curve 5 is situated below the flow rate axis. In this domain the NPSH_R of the system inducer/impeller is largely determined by the $\text{NPSH}_{3,\text{inducer}}$ of the inducer. In the region $Q_u < Q < Q_B$ the control of the NPSH_R of the pump passes from the inducer to the impeller which now obtains insufficient pressure. The operating limit is located to the left of curve 5 and left of the point in which curves 1 and 2 intersect. However, it is not easy to accurately predict the steep rise of curves 2 and 5. In brake operation $Q > Q_B$, curve 5 is located above curve 1. Here the pump cannot be operated under any circumstances because the impeller receives less inlet pressure than without an inducer. The calculated NPSH requirement $\text{NPSH}_{3,\text{inducer}}$ of the inducer is plotted as curve 2 while curve 4 constitutes the measured NPSH_3 of the system inducer plus impeller.

These relationships determine the procedure for calculating the inducer outlet as follows:

9. The measured or calculated $\text{NPSH}_{3,\text{imp}}$ of the impeller (without inducer) is plotted as a function of the flow rate, curve 1.
10. The curve of the required $\text{NPSH}_{3,\text{inducer}} = f(Q)$ of the inducer is estimated according to Eq. (T7.6.10) [30]. Compared with other correlations this relationship has the advantage that it produces a steep rise of the curve with increasing flow rate when the incidence tends to zero (the nominator of the term on the right side of this equation then tends to zero). Since the constant k_3 varies over a wide range, it is recommended to estimate the suction specific speed at the design point from Eq. (7.23a) and use Eq. (7.6.10) only for the shape of the curve $\text{NPSH}_3 = f(Q)$.
11. A blade outlet angle β_{2B} on the outer streamline is selected and the static pressure rise in the inducer calculated. This is done using Eq. (T7.6.15) which can be derived from Table 3.3. The head coefficient is calculated on the mean

streamline. The flow deflection is taken into account by way of the deviation angle using Eq. (T7.6.14). The hydraulic efficiency of the inducer can be estimated from Eq. (T7.6.13). Depending on the flow rate ratio q^* and the geometry, it is in the range of $\eta_h = 0.7$ to 0.9 . For this calculation it would also be possible to use correlations for the head coefficient given in the literature. When applying correlations, it must be considered that individual geometrical parameters such as the hub diameters on the inlet and outlet of the inducer cannot be assessed in their effect on the pressure increase. Furthermore, the geometries of the inducers examined in the literature are not adequately known. In addition, the *static* pressure rise (not the head rise or the total pressure rise) must be used as a criterion since it is solely responsible for the NPSH improvement.

12. The rise in the static pressure caused by the inducer must be calculated for the flow rate range between shockless entry or $H_p = 0$ and the flow rate Q_u where H_p is greater than $NPSH_{3,imp}$ of the impeller alone. At partload the calculation becomes uncertain due to growing inlet recirculation. H_p increases due to the centrifugal effect (streamline displacement) so that roughly linear characteristics $H_p = f(Q)$ are obtained which tend towards the point $\psi_{p(Q=0)} = 1.0$.
13. The difference between $NPSH_{3,La}$ (curve 1) and the static pressure rise H_p (curve 3) gives an indication of the maximum flow rate which is definitely to the left of the resulting curve 5.
14. To facilitate assessing this limit, the curve $NPSH_{3,inducer} = f(Q)$ of the inducer (curve 2) is also taken into account.
15. If the maximum flow rate does not meet the requirements, the calculation according to steps (10) to (13) must be repeated with another outlet angle. With respect to partload operation, over-sizing should be avoided.
16. When the outlet angle has been established, all parameters of the outlet velocity triangle according to Table 3.2 can be determined. To verify the configuration, w_2/w_{1a} should be calculated too. To avoid premature flow separation and efficiency loss, the deceleration ratio w_2/w_{1a} should not fall below the value of 0.7 ("de Haller criterion"). The lift coefficient according to Eq. (T7.5.8) should not exceed the value $\zeta_{a,a} = 0.5$ on the outer streamline.
17. The blade angles on the mean and inner streamlines are selected according to $u \times c_u = \text{constant}$ while a compromise is made on the hub (if required) to avoid excessive blade angles, Chaps. 7.5 and 7.6. The larger the hub diameter, the easier it is to configure this free-vortex blading.

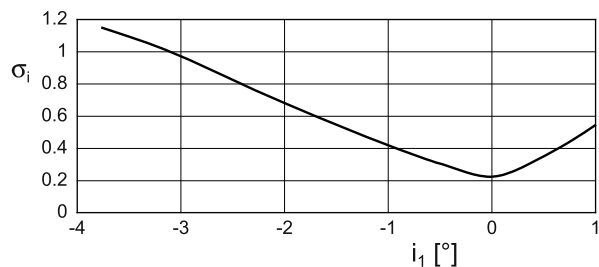
The calculation of the pressure rise in the inducer is quite uncertain since flow deflection and hydraulic losses are affected by intense secondary flows which occur as the result of thick boundary layers and centrifugal forces, Fig. 5.7.

7.7.2 Design and Shaping of an Inducer

After determining the inlet and outlet diameters as well as the blade angles, the inducer can be designed. The design follows Chap. 7.2.2. The following information can help to select appropriate blade shapes and hub contours:

- Blade profile at inlet: The first section of the blade is usually designed wedge-shaped with a leading edge as thin as possible. The minimum required blade thickness is given by blade stresses and natural frequencies. The length L_k of the wedge-shaped portion is between 50 and 80% of the pitch, Fig. 7.33. The wedge angle ϑ_k , which makes up the profiling on the suction surface, is selected as $\vartheta_k = (0.3 \text{ to } 0.5) \times \beta_{1B,a,PS}$. The thin inlet profile is important to obtain good suction capability. It also means low cavitation inception coefficients ($\sigma_i \approx 0.2$ to 0.3) in shockless approach flow. However, σ_i increases rapidly even with a minor incidence, as shown in Fig. 7.36.
- The blade angle β_{1B} is best kept constant over the first part of the blade length, approximately up to $x/L = 0.25$, in order to produce less deflection of the flow and consequently reduce the low-pressure peak and the generation of cavitation bubbles.
- If the pressure distribution is calculated, it should be checked that the blade design ensures that the static pressure in the throat area on the blade suction surface exceeds the vapor pressure, that is $p_{ss(x=t)} > p_v$.
- A conical hub fosters the pressure build-up in the inducer due to centrifugal forces. At the same time the tendency towards flow separation decreases with the amount of hub taper that can be designed.
- The blade lengths are selected in a relatively large range $L/t = 1.5$ to 2.5 . If the overhung of the inducer permits, values in the range of 1.5 to 1.8 should be the target. If the blades are made too short, the pressure rise in the inducer is impaired since the deviation angle increases. Moreover, the suction capability suffers and there is a growing tendency for pressure pulsations at partload. Blades with $L/t > 2.5$ do not seem to bring about any advantages.
- The blade outlet angles on the outer streamline generally are below 20° , but this depends on the inducer and impeller inlet diameters. The blades are slightly tapered towards the outlet.

Fig. 7.36 Visual cavitation inception of an inducer as a function of the incidence



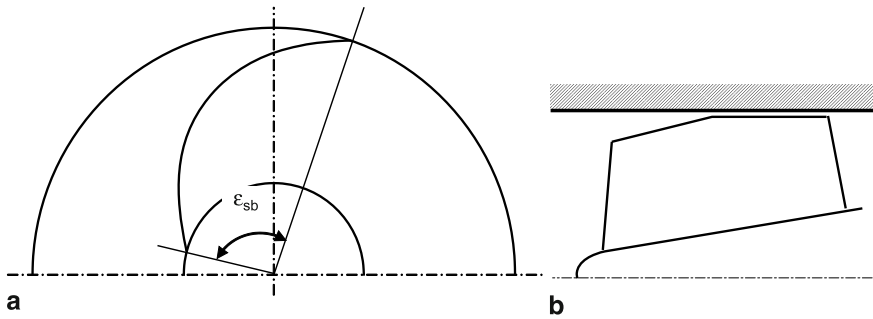


Fig. 7.37 Sweep-back of the inducer blade leading edge

- Efficient inducers are nearly always designed with a “sweep-back” according to Fig. 7.37a. A sweep-back does not only increase the suction specific speed but also moderates the tendency to pressure pulsations. During the tests in [3] the optimum wrap angle ϵ_{sb} (defined according to Fig. 7.37) lay between 65 to 90° on the outer streamline. At half the blade height ϵ_{sb} should be selected roughly half as large as on the outer diameter. Compared with a basic test with $\epsilon_{sb} = 29^\circ$, a suction specific speed increase from $n_{ss} = 280$ to $n_{ss} = 510$ was obtained with optimum sweep-back. The cavity extensions observed were correspondingly smaller and the pressure pulsations were also lower. It should be noted that the blade leading edge in Fig. 7.37 is already strongly inclined towards a radial line on the hub, i.e. it does not join the hub radially.
- The outer contour of the inducer is often tapered at the inlet, Fig. 7.37b. At the inducer inlet there is therefore a large gap between the blades and the casing. This design also reduces the pulsations at low partload and low $NPSH_A$.

7.7.3 Inducer Performance Prediction

If no test data for the blade parameters (λ_w and λ_c) are available for the type of design intended, a statistical correlation can be used to determine the expected n_{ss} and $NPSH_R$ from selected geometric parameters. In order to establish such a correlation, inducer tests from various sources (including [82–85]) were analyzed. The results are shown in Fig. 7.37-1 where the cavitation coefficient is plotted as function of the quantity x_σ which contains the flow coefficient and geometric data. The correlation, Eq. (7.23a) is based on the data near design flow (symbol x), while the graph also shows data at off-design conditions (symbol Δ). Most of the data points are within $\pm 15\%$ of the correlation, while the scatter is larger at off-design operation (not surprisingly). In view of the low absolute $NPSH_R$ the uncertainty of the test data is presumably even greater than $\pm 15\%$. Furthermore, the data from literature are often incomplete and not always easy to interpret. The plot demonstrates

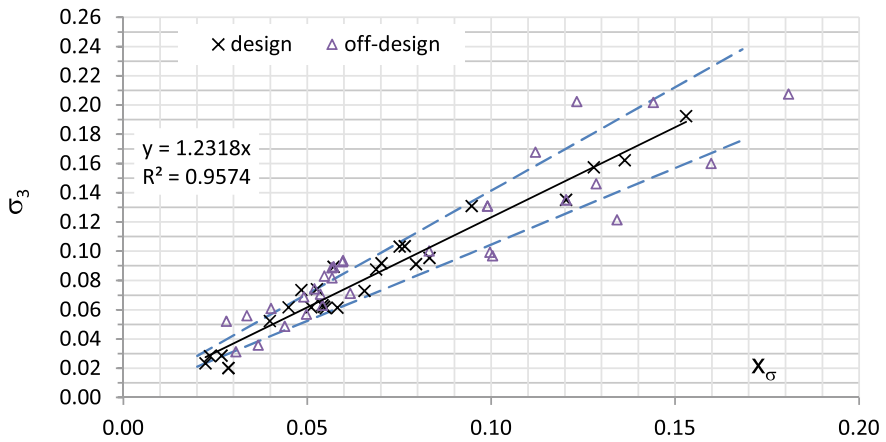


Fig. 7.37-1 Cavitation coefficient for inducers correlated by Eq. (7.23a). The dashed lines mark $\pm 15\%$ of the correlation

that the correlation can also be used to predict the $NPSH_R$ as a function of the flow rate, which is important because the BEP of an inducer is not well defined (the BEP of the pump is given by the volute and the impeller - and with low n_q mostly by the volute). An evaluation of the factor k_3 in Eq. (7.6.10) from the same test data gave no useful correlation.

$$x_{\sigma} = \varphi_1^{0.43} \varphi_{1,SF}^{0.75} \left(\frac{L}{t}\right)_a^{-0.2} \left(\frac{d_{1i}}{d_{1a}}\right)^{0.3} \left(\frac{d_{2a}}{d_{2i}}\right)^{-0.1} \left(1 + \frac{\varepsilon_{sb}}{100}\right)^{-0.47} \left(\frac{z_v - L_a}{3}\right)^{0.24}$$
$$\sigma_3 = 1.26x_{\sigma} \tag{7.23a}$$

For $L/t > 2$, set $(L/t)_{\max} = 2$ in the correlation since blades longer than that are not expected to bring much further improvement. The correlation covers the following range of design data ($n_{ss} = 280$ to 800).

Table D7.3-1 Inducer data to Fig. 7.37-1												
	z_{La}	d_{1a}	d_{1i}/d_{1a}	d_{2i}/d_{1i}	ϕ_1	i_{1a}	$\beta_{1B,a}$	$\beta_{2B,a}$	$(L/t)_a$	ε_B	$\phi_{1,SF}$	$\phi_1/\phi_{1,SF}$
		mm	—	—	—	°	°	°		°	—	
Average			0.27	1.4	0.15	2.8	11	14	2.0	32	0.20	0.78
Maximum	4	300	0.53	2.9	0.30	7.5	19	22	3.3	93	0.34	1.20
Minimum	2	50	0.13	1.0	0.057	−1.4	6.5	8	1.0	0	0.11	0.42

- The main parameters according to the correlation are:
- (1) The inlet flow coefficient φ_1 : Low $NPSH_R$ require low φ_1 .

- (2) The coefficient of shockless entry flow $\varphi_{1,SF} = \tan\beta_{1B,a}/\tau_{1a}$ accounts for the effect of the blade inlet angle at the outer streamline: low $NPSH_R$ requires low $\beta_{1B,a}$ in order to reduce blade loading and cavitation.
- (3) Length to pitch ratio $(L/t)_a$: Longer blades reduce blade loading and tend to reduce cavitation. However, $(L/t)_a > 2.5$ to 3 must not be surpassed. Above $(L/t)_a > 1.5$ to 2 little or no improvement is expected. Even though extra-long blades have little effect on the head produced by the inducer, cavitation bubbles may diminish in long channels.
- (4) Number of blades: reducing the blade number increases the triangular section up to the throat area and reduces blade blockage; both effects reduce the $NPSH_R$. However, the axial length of the inducer increases for a selected ratio $(L/t)_a$. For this reason, 3 blades are often selected even though two blades would promise lower $NPSH_R$.
- (5) Reducing the hub ratio at the inlet, d_{1i}/d_{1a} increases the flow area and reduces $NPSH_R$.
- (6) Reducing the hub diameter d_{2i} at the inducer outlet (increasing the ratio d_{2a}/d_{2i}) seems to be beneficial - maybe due to larger channels.
- (7) Sweep back ε_{SB} : improves $NPSH_R$ and reduces pressure pulsations and instabilities of the cavity flow.

Overall, these parameters and their trends are plausible (in fact most act similar to impellers). Attempts to make a correlation with the ratio $\varphi_1/\varphi_{1,SF}$ failed because the absolute blade angle or stagger angle has a dominating influence on the blade loading, hence the pressure minimum at the leading edge and cavitation.

7.7.4 Inducer Design Systematic

Similar to the concept introduced in Chap. 7.14 for impellers, a design systematic can be developed for a range of inducers. The idea is to use a set of rules and equations by which an inducer can be designed for any given flow rate, speed and inlet diameter. The latter is given by the pump into which the inducer shall be installed. The objectives and concepts are:

- “Design systematic” means that the inducer *geometries are fully described by analytical equations*. This approach will reduce the uncertainty of $NPSH_R$ -prediction although the endeavor is limited by the effect of the impeller downstream of the inducer
- The performance curves are given by: $\sigma_R = f(\varphi_1)$ and $\psi_p = f(\varphi_1)$
- The inducer geometry is fully described in dimensionless terms. Therefore, the inducers can be applied to any pump (independent of n_q) by:
 - Scaling of any inducer selected from the range to any size and speed
 - Develop inducer geometry for any new φ_1 not yet designed

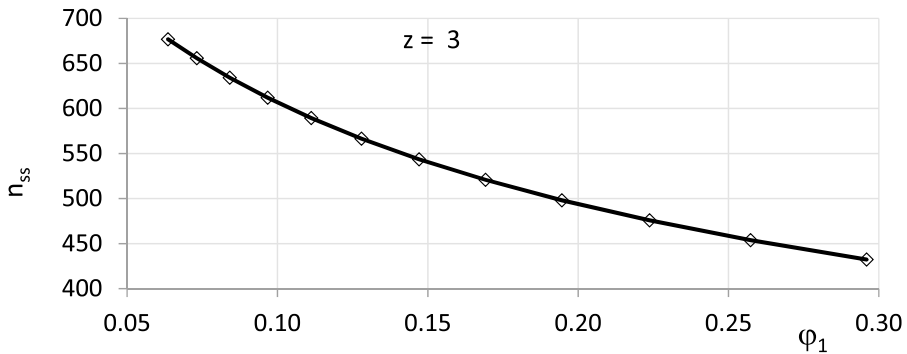


Fig. 7.37-2 Suction specific speed for a series of inducers with 3 blades

- Since the geometry follows from analytical equations, the new geometry is consistent with the design of the adjacent geometries (designed for lower resp. higher flow coefficients)
- The range covers the flow coefficients used for industrial inducers. As an example, Fig. 7.37-2 shows the suction specific speeds as function of the inlet flow coefficient.

For any given application n , Q_{opt} and impeller inlet diameter d_{1a} are specified. Selecting the inducer inlet diameter (equal to d_{1a} or d_{sp} for example) and the hub diameter at the inlet, ϕ_1 and n_{ss} can be calculated. If the n_{ss} is not sufficient, the inducer inlet diameter, must be increased.

7.7.5 Matching the Inducer to the Impeller

Frequently the inducer is mounted upstream of the impeller of a standard pump whose $NPSH_R$ is unable to satisfy the specific requirements of the application. If an impeller of moderate suction capability is combined with an inducer of high suction specific speed, there is, however, no guarantee that the expected reduction of the $NPSH_R$ will actually be achieved. Especially the maximum flow rate can be limited through excessive incidence of the downstream impeller if cavitation occurs on the blade pressure surfaces. To check this, the flow angle β_2 at the inducer outlet is calculated from $\beta_2 = \beta_{2B} - \delta_2$ with δ_2 from Eq. (T7.6.4) (or according to Table 3.2). If β_2 is larger than the blade inlet angle β_{1B} of the downstream impeller, the blade leading edges of the latter can be adjusted to a certain extent by cutting back the blades and profiling them on the pressure surface. To keep the discharge angle as low as possible, the pressure increase in the inducer should not be chosen unnecessarily large.

Figure 7.38 shows measured flow velocities and angles at the outlet of an inducer. Characteristically the meridional velocity near the outer streamline is far

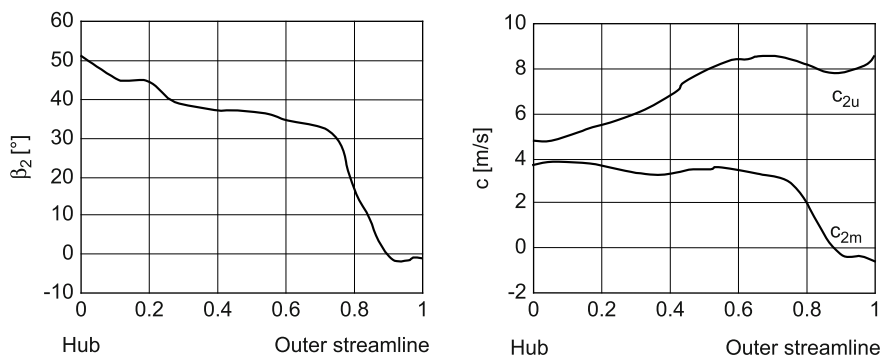


Fig. 7.38 Outlet flow angle and absolute velocity at the outlet of an inducer, Sulzer Pumps

below the mean value while the circumferential velocity is above average. Consequently, the discharge angles are considerably smaller on the outer streamline and far greater near the hub than would be expected from the calculation. These three-dimensional effects are already pronounced in the design point and are reinforced at partload (see also Chap. 5.2 and Fig. 5.7). The deficit in the meridional velocity near the outer streamline is also caused by the tip clearance flow and the boundary layer thickness. With growing tip clearance the deviation angle or the difference between blade and flow angles increases. Owing to the distorted flow conditions the calculation of the flow angles at the inducer outlet becomes quite uncertain.

An acceleration of the fluid from the inducer outlet to the impeller inlet causes a corresponding drop in static pressure and, consequently, an increased tendency to cavitation. The ratio of the velocity w_{1q} in the throat area of the impeller to the approach flow velocity w_1 (which is identical to the relative velocity w_2 at the inducer outlet) should therefore also be verified, Chap. 7.2.1 item (15): at the maximum flow rate $w_{1q}/w_2 = 1$ should not be significantly exceeded.

7.7.6 Recommendations for Inducer Application

The large diameter ratio d_1/d_{1i} causes partload recirculation to start near the design point. At low flow rates recirculation then becomes very intense inducing a strong swirl in the inlet pipes of end-suction pumps. As a consequence of the rotation a parabolic distribution of the static pressure $p(r)$ builds up at the inducer inlet. At low suction pressures a vapor-filled vortex can be generated in the center of the suction pipe. As discussed in Chap. 6.5.1, such vortices can cause severe pulsations. By a suitable design of the inducer it is possible to largely avoid such kinds of pulsations. Favorable in this respect seem to be: long blades (high solidity L/t), variable pitch design, sweep-back and a tapered outer contour of the inducer. However, it is not yet

possible to provide generally valid design criteria for mastering this highly three-dimensional two-phase flow. For this reason it is necessary to prove by way of experiment that no impermissible pulsations and excitation forces occur in the demanded operating range. To this end, the water in the test circuit has to be largely de-aerated since the pulsations are greatly attenuated in air-saturated water. If at all possible, the test should be done with the same circumferential velocity as in the planned operation.

As a rule, pumps for industrial application are required to operate also at low partload. To be able to satisfy this requirement without excessive recirculation, the suction specific speeds of industrial inducers are limited to approximately $n_{ss} = 500$ to 700. If this range is markedly exceeded, the $NPSH_3$ at partload increases so that the inducer may no longer satisfy its actual purpose of lowering $NPSH_3$ throughout the operating range.

Obviously the target should be to restrict the partload operation as far as possible and to entirely avoid operation at $q^* < 0.3$ (except during start-up).

Major cavitation zones are created in the inducer in most applications. Despite the low $NPSH_A$ - i.e. low implsion pressure - the implsion energy can therefore be considerable. While hardly any cavitation damage is to be expected when pumping hydrocarbons, the circumferential velocities of the inducers when pumping water have to be limited in order to reduce the risk of cavitation damage (especially with de-aerated cold water, refer also to Table 6.2). For pumping water it is therefore advisable to use high-alloy steels with suitable cavitation resistance. The limit for the circumferential velocity cannot be determined generally; it may be within a range of 25 to 30 m/s when pumping water and 35 to 40 m/s with hydro carbons. These limits can be inferred from published applications, e.g. [33]. Initial cavitation bubbles (corresponding to $NPSH_i$) are created in the gap between casing and inducer.

Inducers are able to some extent to transport gas-laden liquids. Inducers combined with closed impellers are claimed to handle gas fractions of up to 25% of the flow rate at the inducer inlet; combined with semi-open impellers 35% and with open impellers even 40%. The gas-handling capabilities depend on the geometry and should be confirmed by testing, see also Chap. 13.2.

The Q-H-curve and the efficiency virtually do not change near the best efficiency point when installing an inducer. Due to the energy transfer in the inducer the circumferential component of the absolute velocity at the inducer outlet is increased, but the downstream impeller has to perform correspondingly less work in accordance with the Euler equation, Eq. (T3.3.1). The efficiency of the pump tends to fall since the hydraulic losses in the inducer are proportionally slightly higher than in the impeller, but this effect is sometimes hardly measurable.

However, an inducer increases the shut-off head by several percent due to the intense inlet recirculation and the associated rise of the centrifugal head component, Chap. 5.4.1. In the overload range the head drops slightly compared with the pump without inducer but this is likely to become noticeable only when the inducer operates in the range $H_p < 0$ ($Q > Q_B$), Fig. 7.35.

Inducers have relatively thin blades with a large blade height. When the inducers are cast possible blade deformations must be monitored. In particular it should be

Table 7.7 Diffuser and volute design calculations			Eq.
Given	n, Q _{opt} , H _{opt}		
Diffuser or volute design flow rate	Q _{Le} = Q _{opt} + Q _{s3} + Q _E		7.7.1
Circumferential component of absolute velocity at impeller outlet	c _{2u} = $\frac{g H}{\eta_h u_2} + \frac{u_{1m} c_{1u}}{u_2}$		7.7.2
Circumferential component of absolute velocity at diffuser inlet	c _{3u} = $\frac{d_2 c_{2u}}{d_3}$		7.7.3
Meridional component of absolute velocity at diffuser inlet. For volutes set: τ ₃ = 1.0	c _{3m} ' = $\frac{Q_{Le} \tau_3}{\pi d_3 b_3}$		7.7.4
Flow angle at diffuser inlet with blockage	tan α ₃ ' = $\frac{c_{3m}'}{c_{3u}}$		7.7.5
Diffuser vane inlet angle	α _{3B} = α ₃ ' + i ₃	incidence i ₃ = ±3°	7.7.6
Throat area of a volute of circular section or equivalent circle of a volute with an arbitrary section	X _{Sp} = $\frac{Q_{Le}}{\pi c_{2u} r_2} \frac{\epsilon_{sp}}{2\pi}$		7.7.7
	d _{3q} = X _{Sp} + √2 d _z ' X _{Sp}	A _{3q} = $\frac{\pi d_{3q}^2}{4}$	
Factor for diffusers or volutes with rectangular cross section	X _{Le} = exp { $\frac{Q_{Le}}{b_3 c_{2u} r_2 z_{Le}}$ }		7.7.8a
Outer radius r _a of the end section (throat area) of a rectangular volute	r _a = r _z ' X _{Le}		7.7.8
Height of the end section (throat area) of a rectangular volute	a ₃ = r _a - r _z ' = r _z ' exp { $\frac{Q_{Le}}{b_3 c_{2u} r_2} \frac{\epsilon_{sp}}{2\pi} - 1$ }		7.7.9
Diffuser inlet width f _{a3} = 1.1 to 1.3	a ₃ = f _{a3} $\frac{d_3}{2}$ { X _{Le} - 1 }		7.7.10
Diffuser inlet width calculated for a selected deceleration ratio c _{3q} /c ₂	a ₃ = $\frac{Q_{Le}}{z_{Le} b_3 \sqrt{c_{2m}^2 + c_{2u}^2}} \left(\frac{c_2}{c_{3q}} \right)$		7.7.11
Return channel outlet width	b ₆ = $\frac{Q_{Le}}{\pi d_6 c_{6m}}$	c _{6m} = (0.85 to 0.9) c _{1m}	7.7.12
Meridional component of absolute velocity at return channel inlet	c _{5m} = $\frac{Q_{Le}}{\pi d_5 b_5}$		7.7.13
Circumferential component of absolute velocity at return channel inlet	c _{5u} = $\frac{c_{4u} r_{4m}}{r_{5m}}$		7.7.14
Flow angle at return channel inlet	α ₅ = arctan $\frac{c_{5m}}{c_{5u}}$		7.7.15
Gap A and overlap x _{ov} (Fig. 9.1)	$\frac{\text{gapA}}{r_2} = 0.007 \text{ to } 0.01$	$\frac{x_{ov}}{\text{gapA}} = 2 \text{ to } 4$	7.7.16
Gap B	s. Table 10.2		
ε _{Sp} is given in radians; for volutes with equal spacing: ε _{Sp} = 2 π/z _{Le}			

checked that the desired blade outlet angles are achieved: if β_{2B} is too small, the necessary pressure increase is not attained and the steep rise of the NPSH occurs at too low a flow rate. The thin blades also require a verification of the natural frequencies and stresses in order to prevent blade fractures.

When installing an inducer in an end-suction pump, the impeller overhung increases. This has to be considered when evaluating the shaft deflection and the critical speed. Also some additional excitation forces in radial direction must be expected. Radial excitation forces can (for instance) be created by the tip clearance flow or through periodically variable flow conditions that might develop in the channels of the inducer as the result of varying cavity volumes, [63].

7.8 Volute Casings

7.8.1 Calculation and Selection of Main Dimensions

In the volute casing the kinetic energy present at the impeller outlet is converted into static pressure with as few losses as possible. The volute casing then directs the fluid into the discharge nozzle or, in multistage pumps, into the following stage. Prior to calculating a volute, the boundary conditions for the casing design must be established. In particular it must be decided whether a single or double volute is required. The selection criteria are:

- **Single volutes** are the least expensive solution in terms of manufacturing costs; they are best accessible for the dressing of the cast channels. Their disadvantage consists in considerable radial forces which develop during operation at off-design conditions according to Chap. 9.3 through disturbances of the circumferential symmetry of the flow in the casing. The radial forces generate bearing loads, bending stresses in the shaft and shaft deflection which can threaten the reliability of the machine. The heads up to which single volutes can be sensibly used depends on the specific speed, the design of the pump (especially of the bearing housing), the shaft thickness and the bearing span or the overhung of the impeller. When pumping water, the limit is about $H_{opt} = 80$ to 120 m at $n_q < 40$. At high specific speeds a double volute may be indicated already for heads above $H_{opt} = 60$ to 80 m. When pumping liquids with densities considerably lower than water, the limit is correspondingly higher since the radial forces are proportional to $\rho \times g \times H$.
- **Double volutes** are employed when bearing loads, shaft stresses and shaft deflection become impermissibly high without measures for radial thrust reduction so that their control would require an excessively high design effort and cost. The rib between the inner volute and the outer channel also reduces the casing deformation under internal pressure, which facilitates the casing design at high specific speeds. The pressure to be applied during the hydrostatic test of the casing must also be considered.